

README.md

Kanokpon Mettasat

Electrical & Computer Engineering, Honors Program · Virginia Military Institute, Class of 2027 · Overall GPA 3.79 / Major GPA 4.00 -- Institute Honors Program

GPA: overall 3.6 · major 4.00

sections

**about.md**

Who I am, one paragraph.

**education/**

VMI · CU Boulder · RTA Cadet.

**projects/**

Dashboard · LINE bot · Quant research.

Audience-specific: [recruiters](#) | [interns](#) | [grad_school](#)

More: [notes](#) | [reading_list](#)

activity.json

contact

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Electrical & Computer Engineering, Honors Program · Virginia Military Institute, Class of 2027 · Overall GPA 3.79 / Major GPA 4.00 -- Institute Honors Program

Originally a **Royal Thai Army cadet (Class 74)**, selected first in the cohort for the army's study-in-US program in 2022. After completing the University of Colorado Boulder's International English Center (Levels 4 through 7) in 2022-2023, enrolled at Virginia Military Institute reading Electrical and Computer Engineering. Beyond coursework, builds software (FastAPI services, LINE OA bots) and runs quantitative trading research as an independent practice -- pre-registered hypotheses, walk-forward out-of-sample validation, and explicit bias auditing.

about/skills.json

LANGUAGES			WEB / API		
Python	JavaScript (ES2020)	TypeScript	FastAPI	Jinja2	HTML5 / CSS3
PowerShell	Bash				REST
DATA / QUANT			TOOLING		
pandas	NumPy	MetaTrader 5 API	pytest	pyright	Playwright
Probabilistic Sharpe / Deflated Sharpe			GitHub Actions	Git	
INFRASTRUCTURE					
Windows Server	Tailscale Funnel	Scheduled Tasks			
SQLite	LINE Messaging API				

Virginia Military Institute

Aug 2023 -- May 2027 (expected)

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  "institution": "Virginia Military Institute",
  "location": "Lexington, VA",
  "program": "BS, Electrical and Computer Engineering",
  "dates": "Aug 2023 -- May 2027 (expected)",
  "details": [
    "Institute Honors Program, VMI (selected Jan 2024) -- Honors scholarship since Jul 2024, USD five thousand per academic year",
    "Dean's List, multiple terms",
    "Overall GPA 3.79 / Major GPA 4.00",
    "Term GPA 4.00 (Year 2 Term 1)",
    "Term GPA 4.00 (Year 2 Term 2, all-A semester)",
    "Term GPA 3.81 (Year 3 Term 1)",
    "Term GPA 3.70 (Year 3 Term 2)",
    "Tau Beta Pi -- Engineering Honor Society, Virginia Chapter (inducted Nov 2025)",
    "IEEE SoutheastCon 2026 -- 4th place, Undergraduate Circuit Design (Alabama, Mar 2026)",
    "Tau Beta Pi Conference -- VMI representative (Mar 2026)",
    "OGA Representative, Hotel Company -- oversees new-cadet orientation (May 2026)",
    "Cadre Old Corps Corporal + Cadre Rat Challenge + MathLab Tutor (Year 2 onward)",
    "Expert classification, Rifle (May 2025)",
    "Commendation from the Chief of Defence Forces, US Reserve Command HQ (Apr 2024)",
    "Community service in Puerto Rico with the VMI Glee Club (Mar 2024)",
    "SAT 1260 (March 2023)"
  ]
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```
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  "location":   "Boulder, CO",
  "program":    "International English Center, Levels 4-7",
  "dates":      "Aug 2022 -- May 2023",
  "details": [
    "Certificate of completion, English Programs"
  ]
}
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Royal Thai Army Cadet School (Chulachomklao) -- Class 74

```
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  "location":    "Thailand",
  "program":     "Army cadet, English Program (EP)",
  "dates":       "Mar 2022 -- Aug 2022 (pre-US dispatch)",
  "details": [
    "Selected #1 of 348 Army cadets for the army's study-in-US programme (May 2022)",
    "Section Leader, Section A.1 (EP)",
    "Dispatched on a 5-year Royal Thai Army scholarship to VMI (Aug 2022 -- May 2027)"
  ]
}
```



Honors, awards, leadership (12)

2021 -- 2026

2026-05

OGA Representative, Hotel Company -- new-cadet orientation lead

@ VMI

2026-03

IEEE SoutheastCon 2026 -- 4th place, Undergraduate Circuit Design

@ Alabama

2026-03

Tau Beta Pi Conference -- VMI representative

@ Tau Beta Pi Society

2025-11

Tau Beta Pi -- Engineering Honor Society, Virginia Chapter

@ VMI

2025-05

Expert classification, Rifle

@ VMI

2024-07

VMI Institute Honors Program scholarship (USD five thousand / year)

@ VMI

2024-04

Commendation from the Chief of Defence Forces

@ US Reserve Command HQ

2024-01

Selected into the VMI Institute Honors Program

@ VMI

2022-05

Selected #1 of 348 for the army study-in-US programme

@ Royal Thai Army Cadet School

2022-02

AFAPS completion + ALCPT silver medal (Commander-in-Chief)

@ AFAPS

2021-12

Honorable Mention, 20th National Physics Olympiad

@ Thailand

2021-07

Vice-President, Army cadet section + Honor Council

@ AFAPS



2024-2026 update **Year 2 (VMI, Aug 2024 -- May 2025)**. Coursework in the ECE core: Digital Systems Design, Electrical Circuit Analysis, Discrete Mathematics, multivariable Calculus. Closed Term 2 with a Term GPA of 4.00 (all-A semester). **Summer 2025 (Bangkok)**. Returned home; spent the off-term refining the personal quantitative-trading research workflow (multi-asset OHLCV in `pandas`, walk-forward out-of-sample validation, pre-registered hypotheses). **Year 3 Term 1 (VMI, Aug 2025 -- Dec 2025)**. Computer Architecture, Microcontrollers, Electronics, Modern Physics. Term GPA 3.81. **Tau Beta Pi** (Engineering Honor Society, Virginia Chapter) invitation in Nov 2025. **Year 3 Term 2 / Spring 2026 in progress**. Honors-program coursework; competed at **IEEE SoutheastCon 2026** in Alabama and placed **4th in Undergraduate Circuit Design** (Mar 2026). Represented VMI at the **Tau Beta Pi national conference** the same month. ### Software shipping cadence (research-only, public-repo dates) - **May 2026 -- LINE OA personal-secretary bot**. Multi-owner LINE Messaging API service with daily 18:00 BKK eve-of briefing, 1-hour pre-alerts, Sunday weekly digest, daily SQLite snapshot backup, Telegram-style task commands (`/list` , `/done` , `/retry`). Deployed unattended on a Windows VPS behind Tailscale. ~201 tests. - **May 2026 -- Trading-research dashboard**. `FastAPI` + Jinja2 monitoring site with multiple pages: hub with a BKK/NY clock tower, project board with phase-locked Gantt rows, 30-agent organisation chart, pixel-art floor-plan view of active agents, and a decision-rules anchor page. Hard-restart deploy via PowerShell; reachable on a Tailscale Funnel URL. ~300+ tests. - **May 2026 -- /portfolio public personal-introduction page**. This page. `FastAPI` + Jinja2 with an IDE-shell layout, multi-route split, view-source modal, Cmd-K command palette, keyboard-driven `vim`-style navigation, and a 365-day commit-activity grid. ~120 tests on this surface alone. ### Quantitative trading research Continued the multi-asset backtest framework through 2026. Current state (per the project's own status doc): - The single primary system in active research is **11-edge FLAT** -- eleven equal-weight signals across XAUUSD, GBPUSD, EURJPY, GBPJPY, USDJPY, US30, US100, ETHUSD with a UNION(Co, C7) selector and a flat (no-pyramid) ladder. Backtest Sharpe ~2.59 / lower-bound Sharpe ~2.27 . - Decision principle is **Sharpe first, leverage last**: rank configurations by the block-bootstrap 5th-percentile Sharpe and pass three probability-of-edge gates (`SR_LB >= 1.0` , `PSR >= 0.95` , `DSR >= 0.95`) before any deployment label. Earlier high-CAGR ladders were retired after independent audits showed them tail-fragile. - The forward-validation window runs on a demo broker via MetaTrader 5; a separate signed spec covers an isolated force-test on a small evaluation account. No claim of a profitable live edge -- the entire framework is research, code, and documentation. A companion literature review followed a **PRISMA-style protocol** across **79 IEEE papers** (May 2026), with a multi-dimensional taxonomy used to guide the 2026 research backlog.

AI OFFICE

วันนี้ใครอยู่ออฟฟิศ

HUB

11-EDGE

ORG TREE

PROJECTS

STOP

BUILD 28268525-1424

เลขา

1 ในห้อง

SYS_READY

ระบบพร้อมใช้งาน

ALL SYSTEMS OK

Trading

6 ในห้อง

TX OK - PF 2.82

ความ SUSURI

5 ในห้อง

PRISMA 89/68/79

DevTech

6 ในห้อง

tests 104/104 PAS

UX/Design

4 ในห้อง

render ok - G6fp

ชีวิต/สุขภาพ

5 ในห้อง

VISA 5/5 docs

กลยุทธ์ วิสัยทัศน์

3 ในห้อง

phase 1 - 1.13

เครื่องมือออฟฟิศ

38 คน

น้องคำ

design · fresh

น้องหน้า

design · fresh

พี่สี่

design · fresh

น้องดีไซน์

design · fresh

พี่มิ่ง

dev · fresh

พี่วิว

dev · fresh

น้องเค

dev · fresh

พี่ออปส์

dev · fresh

เสี่ยวโค้ด

dev · fresh

คุยกับ พี่ธรา

ล้างประวัติ

พิมพ์แล้ว Ctrl+Enter ส่ง...

0 / 500

ส่ง

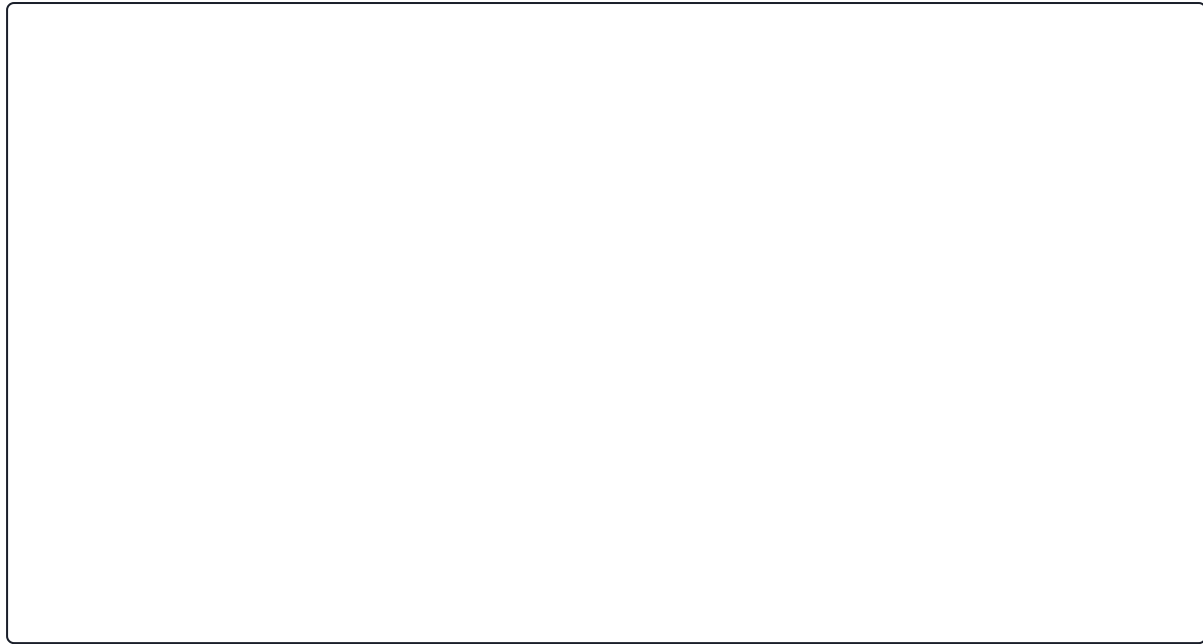
Trading-research dashboard

FastAPI + Jinja2 monitoring and decision-support dashboard for a personal quantitative-trading workflow. Surfaces include project tracking, an editorial organisation chart, a Phaser 3 isometric office-RPG view of active research agents, an 11-edge live-demo monitor with a tail-concentration auditor, and a Smart Money Radar that scores 13F / Form-4 / political-disclosure signals (1-100). Deployed via Tailscale Funnel to a Windows VPS with a hard-restart deploy script.

Python FastAPI Jinja2 Phaser 3 pytest PowerShell

// ~679 tests, 12+ public/private pages, zero-downtime redeploy

[Read deep-dive →](#)



LINE Official Account bot

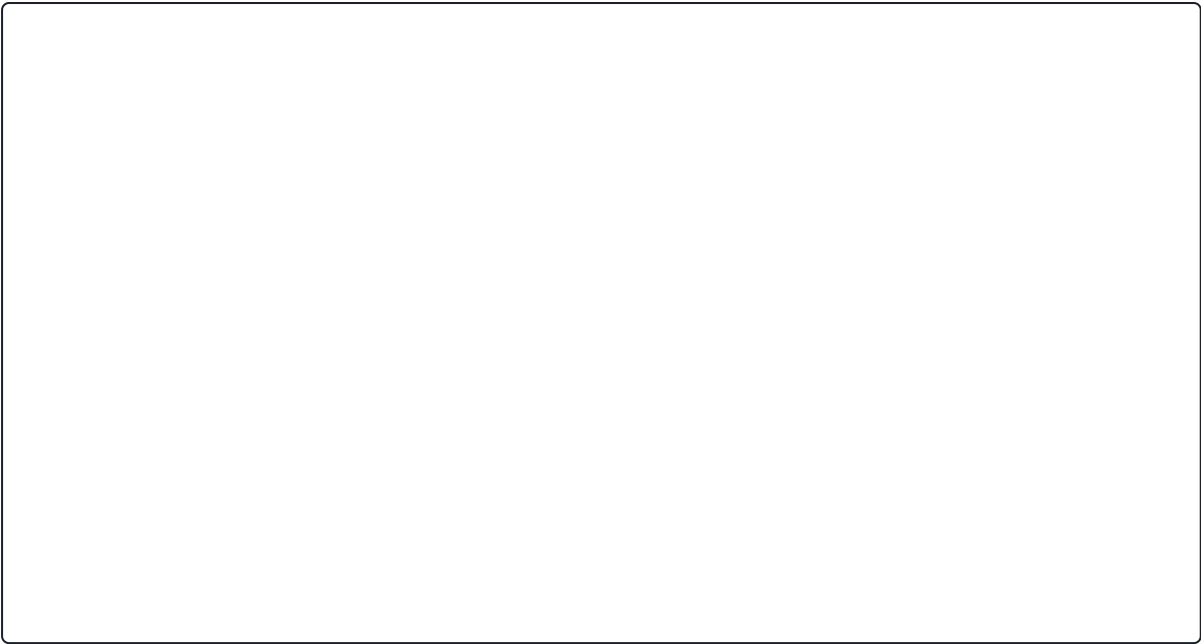
Multi-owner LINE Messaging API bot with scheduled reminders, daily SQLite snapshots, Sunday digest, and pre-alert dedupe. Designed to run unattended on a Windows VPS with a self-healing restart loop.

Python FastAPI SQLite LINE Messaging API

// ~201 tests, multi-tenant

[Read deep-dive →](#)





Quantitative trading research framework

60+ Python backtests across multi-asset OHLCV (FX majors, FX crosses, metals, indices, crypto) with pre-registered hypotheses, walk-forward out-of-sample validation, explicit bias auditing, and probabilistic Sharpe / deflated Sharpe gates. Companion literature review followed a PRISMA-style protocol across 79 IEEE papers with a multi-dimensional taxonomy.

Python pandas MetaTrader 5 API pytest
// 60+ backtests, PRISMA review of 79 IEEE papers
[Read deep-dive →](#)

SURI -- Systematic literature review (IEEE, PRISMA-style)

A pre-registered, PRISMA-style systematic review of seventy-nine IEEE papers on adversarial threats to machine-learning systems, organised by a seven-role taxonomy and a coverage matrix that maps each paper to the role(s) it plays in a defender's mental model. Companion to the quant-research framework: the same discipline applied to the literature side of the work.

IEEE Xplore PRISMA protocol Multi-dimensional taxonomy Coverage matrix
// 79 papers, 7-role taxonomy, single-coder + audit pass
[Read deep-dive →](#)

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+-----+

